25/3
26/3 } FINIRE programma 1° parziale → vettori, Geom. analitica in $\mathbb{R}^3$, $\mathbb{C}$

27/3 11-13 Tutoraggio Aula Z

⟹ Merc. 3/4 Esercizi di revisione
Ven 5/4 SIMULAZIONE 1ª prova

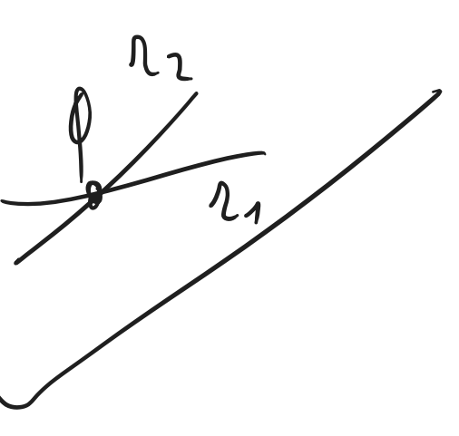
1ª parziale 17/4 — PARI ore 11.15 — DISPARI 12.30

!! Settimana [9/4, 10/4, 12/4] lezioni regolari (CONTENUTI NON INSERITI 1° PARZIALE)

Siano r_1, r_2 2 rette distinte in $\mathbb{R}^3$. Possono essere

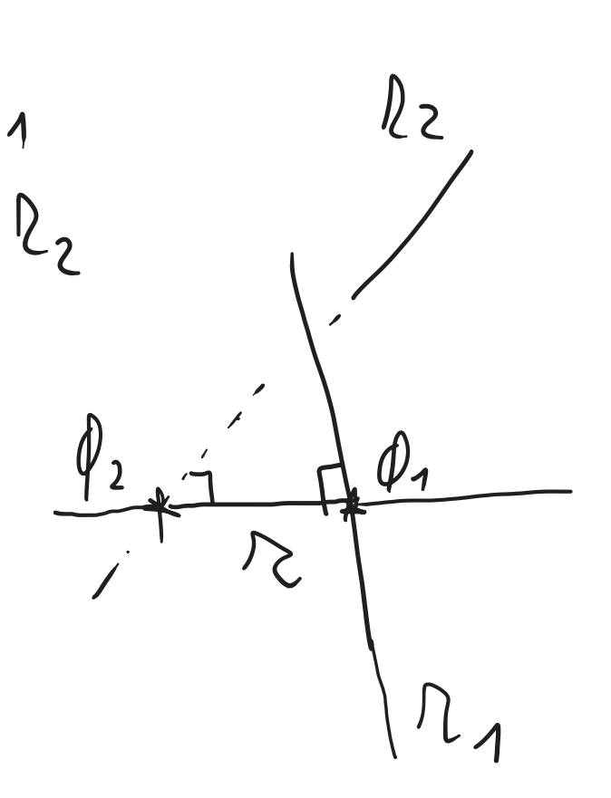
(1) Parallele

(2) incidenti ✓



$P_1 = r \cap r_1$
$P_2 = r \cap r_2$

(3) sghembe (NÉ // né incidenti)



↗ $\exists\, r$ (comune ⊥ a r_1 e r_2) t.c.

(a) $r \perp r_1$, $r \perp r_2$

(b) r è incidente a r_1, e r è incidente a r_2

OSS. $\mathrm{dist}(P_1, P_2)$ rappresenta la minima distanza tra un punto di r_1 e un punto di r_2

Es. $r_1: \begin{cases} x-y-1=0 \\ z-1=0 \end{cases}$ $r_2: \begin{cases} x+z=0 \\ y-1=0 \end{cases}$

Verificare che r_1 e r_2 sono sghembe e determinare la loro comune perpendicolare r.

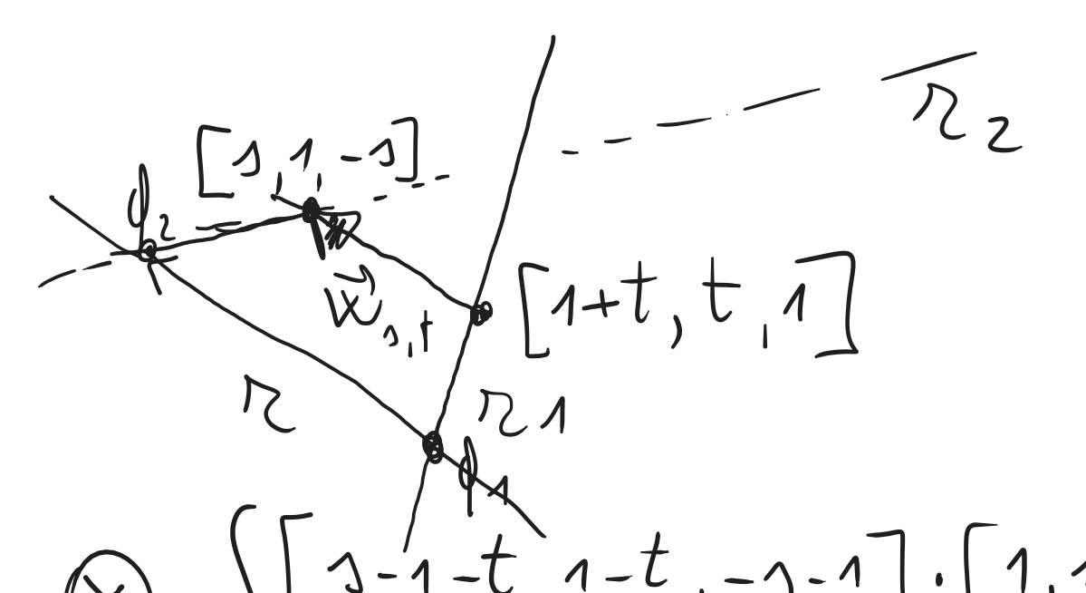
$r_1: [1+t, t, 1]$ $r_1 // \vec{w}_1 = [1,1,0]$

$r_2: [s, 1, -s]$ $r_2 // \vec{w}_2 = [1,0,-1]$

$\vec{w}_1$ NON // $\vec{w}_2$ ⟹ r_1 e r_2 NON sono //

r_1, r_2 saranno incidenti se $\exists\, s,t$ t.c. $\begin{cases} 1+t=s \\ t=1 \\ 1=-s \end{cases}$ $\begin{cases} 2=-1 \\ t=1 \\ s=-1 \end{cases}$ IMPOSSIBILE

Possiamo costruire la loro comune ⊥ r ⟸ SGHEMBE ⟸



$\vec{w}_{s,t} = [s-1-t, 1-t, -s-1]$

Dobbiamo imporre $\begin{cases} \vec{w}_{s,t} \cdot \vec{w}_1 = 0 \\ \vec{w}_{s,t} \cdot \vec{w}_2 = 0 \end{cases}$ ⊛

⊛ $\begin{cases} [s-1-t, 1-t, -s-1]\cdot[1,1,0] = 0 \\ [s-1-t, 1-t, -s-1]\cdot[1,0,-1] = 0 \end{cases}$ $\begin{cases} s-1-t+1-t=0 \\ s-1-t+s+1=0 \end{cases}$ $\begin{cases} s-2t=0 \\ 2s-t=0 \end{cases}$

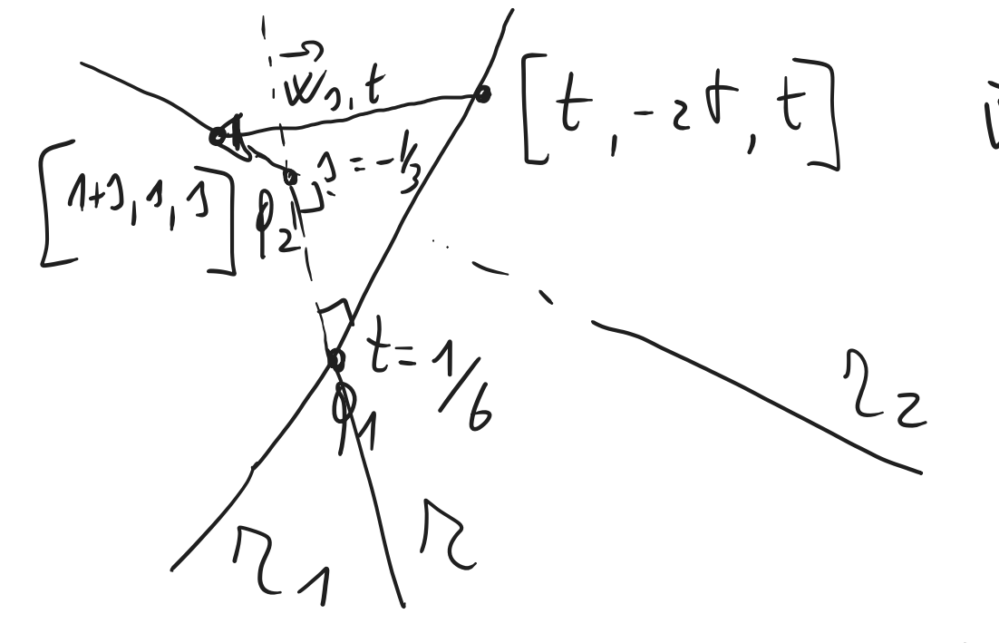
$\begin{cases} s=2t \\ 3t=0 \end{cases} \Rightarrow \begin{cases} s=0 \\ t=0 \end{cases}$ $P_1\,(t=0)$: $P_1 = [1,0,1]$
$P_2\,(s=0)$ $P_2 = [0,1,0]$

$r // \vec{w}_{0,0} = P_2 - P_1 = [-1,1,-1]$

Conclusione: $r: \begin{cases} x=-t \\ y=1+t \\ z=-t \end{cases}$ $r: \begin{cases} x-z=0 \\ x+y-1=0 \end{cases}$

Es 2 $r_1: \begin{cases} x+y+z=0 \\ x-z=0 \end{cases}$ $r_2: \begin{cases} x=s+1 \\ y=s \\ z=s \end{cases}$ $r_2 // \vec{w}_2 = [1,1,1]$

$r_1: [t, -2t, t]$ $r_1 // \vec{w}_1 = [1,-2,1]$ (verificare che sono sghembe)



$\vec{w}_{s,t} = [1+s-t, s+2t, s-t]$ Dobbiamo imporre $\begin{cases} \vec{w}_{s,t} \cdot \vec{w}_1 = 0 \\ \vec{w}_{s,t} \cdot \vec{w}_2 = 0 \end{cases}$ ⟺

$\begin{cases} 1+s-t-2s-4t+s-t=0 \\ 1+s-t+s+2t+s-t=0 \end{cases}$ ⟺

$\begin{cases} t = 1/6 \\ s = -1/3 \end{cases}$ $P_1 = \left[\frac{1}{6}, -\frac{2}{6}, \frac{1}{6}\right]$ $P_2 = \left[\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}\right]$

La comune ⊥ r è la retta che passa per P_1 e P_2

OSS. $r // \vec{w}_{-\frac{1}{3},\frac{1}{6}} = \left[\frac{1}{2}, 0, -\frac{1}{2}\right] // [1,0,-1]$

$r: \begin{cases} x = \frac{1}{6}+t \\ y = -\frac{2}{6} \\ z = \frac{1}{6}-t \end{cases}$ $\begin{cases} 3x+3z-1=0 \\ 3y+1=0 \end{cases}$

$\begin{cases} r // [l,m,n] \\ r \text{ passa per } x_0, y_0, z_0 \end{cases}$ $\begin{cases} \underbrace{\dfrac{x-x_0}{l} = \dfrac{y-y_0}{m}}_{\pi} = \dfrac{z-z_0}{n} \end{cases}$ (π')

Nel nostro caso: $[l,m,n] = [1,0,-1]$ $[x_0,y_0,z_0] = P_1 = \left[\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right]$

$$r: \underbrace{\frac{x-\frac{1}{6}}{1} = \frac{\boxed{y+\frac{1}{3}}}{\underset{?}{0}} = \frac{z-\frac{1}{6}}{-1}}$$ $\begin{cases} y+\frac{1}{3}=0 \\ 3x+3z-1=0 \end{cases}$

Prodotti NOTEVOLI

$a^2-b^2 = (a+b)(a-b)$

$a^n-b^n = (a-b)\underbrace{\sum_{j=1}^{n}\left(a^{n-j}b^{j-1}\right)}$ ⟵

$= (a-b)\left(\underbrace{a^{n-1}+a^{n-2}b+\ldots+ab^{n-2}+b^{n-1}}\right)$

$= a^n + \cancel{a^{n-1}b} - \cancel{a^{n-1}b}$ ~~~~~~ $-b^n$

$a^n+b^n = (a+b)\underbrace{\sum_{j=1}^{n}(-1)^{j-1}a^{n-j}b^{j-1}}$

n dispari

Coefficiente binomiale:

$(a+b)^n = \ldots$

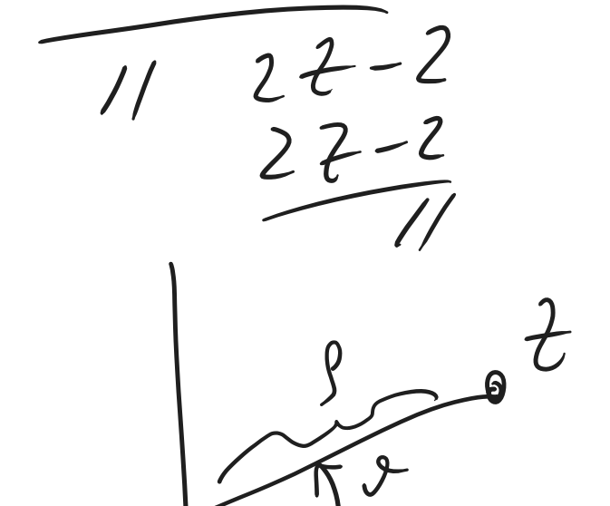
$P(z) = z^4+z^3-z^2+z-2$ $z_0 = 1$ $m_a(z_0) \overset{?}{=}$

$z^4+z^3-z^2+z-2$ | $z-1$
z^4-z^3 | $\underbrace{z^3+2z^2+z+2}$
// $2z^3-z^2+z-2$
$2z^3-2z^2$
// z^2+z-2
z^2-z
// $2z-2$
$2z-2$
//

z_0 NON è radice

$P(z) = (z-1)\,(\underbrace{z^3+2z^2+z+2})$

⟹ $m_a(z_0) = 1$



→ $z = \rho e^{i\theta}$